Gender Differences in Employment During COVID-19 School Closures: Evidence from British Columbia

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1 Introduction

The COVID-19 pandemic created global disruptions, including unprecedented public health measures such as widespread childcare closures and school shutdowns. These measures posed significant challenges for households, through job and economic uncertainty. Parents were hit particularly hard, as the pandemic impacted the balance between work and family responsibilities as a result of school closures.

COVID-19 was officially declared a pandemic by the World Health Organization on March 11, 2020, creating rapid changes to daily life and magnifying pre-existing social and economic inequalities (Blundell et al., 2022). In response, British Columbia's provincial authorities implemented a suite of interventions, including lockdowns and business restrictions, to contain the virus's spread. Just six days later, on March 17, 2020, the BC Ministry of Education mandated the closure of all K–12 schools, disrupting childcare arrangements for families with school-aged children.

However, the policy environment was not uniform. Unlike schools, daycares and early-learning centres were never formally required to close, and beginning in May 2020, limited in-person instruction reopened for the children of essential service workers. This staggered and uneven implementation created meaningful variation in childcare access across households, generating conditions that allow for the examination of how differential exposure to childcare disruptions affected parental labour market outcomes.

Given the historically gendered division of caregiving labour and well-documented disparities in labour market trajectories between mothers and fathers, these pandemic-induced shocks had the potential to further widen gender employment gaps. Reduced access to childcare typically increases unpaid care and domestic responsibilities, adjustments that empirical evidence suggests fall disproportionately on women. This paper therefore investigates whether COVID-19–related childcare and school disruptions exacerbated gender differences in employment among parents in British Columbia, with a focus on how the employment trajectories of mothers and fathers diverged following the onset of these policy measures.

Understanding these dynamics is essential for designing equitable and resilient public policy. By identifying how variation in childcare access shaped parental labour supply during the early stages of the pandemic, this analysis contributes to broader discussions on the role of childcare infrastructure, persistent gender inequalities in the labour market, and households' capacity to adapt to sudden and prolonged shocks. The findings also offer insights into how future crises may propagate through families' time-use constraints, while highlighting the importance of childcare systems in maintaining labour force attachment, particularly for the maternal labour force.

2 Literature Review

The economic ramifications of school closures, especially concerning gender-differentiated employment outcomes, have become a focal point of research. Studies consistently demonstrate that women, particularly mothers, experience disproportionately negative impacts on their employment and work hours.

Many studies have documented pervasive gender inequalities in employment and house-

work that existed prior to the pandemic (Beaujot et al., 2017; Fuller & Cooke, 2018).Specifically regarding COVID-19, studies on the gendered effects in Canada have largely focused on the impacts of the pandemic as a whole on the female labour force. Qian and Fuller (2020) provide an early Canadian benchmark. Using the Labour Force Survey, they document widening gender gaps in employment among parents between February and May 2020 using a logistic regression. Several other studies describe pronounced employment reductions for women in major Canadian urban areas, particularly resulting from the onset of school closures (Singh et al., 2022). Similar studies have been conducted in other countries, namely the US (Collins et al., 2021; Couch et al., 2022), concluding that women, particularly those with young children, disproportionately worked less with the onset of COVID. In the UK, Zamberlan et al. (2021) find that unpaid labour falls upon females in households, even accounting for situations where the woman is the "breadwinner".

Internationally, work exploiting variation in school modality and childcare policy finds that remote schooling or childcare and school closures are associated with larger reductions in mothers' employment and work hours relative to fathers (Garcia & Cowan, 2022; Petts et al., 2021). Some studies qualitatively highlight how women disproportionately absorbed additional unpaid childcare and household labour during closures (Del Boca et al., 2020). Some take a different turn and analyze the impact of return to school on women's employment status. Beauregard et al. (2022) exploit variation in school reopenings in Quebec to estimate the impact of school reopenings on maternal employment and work hours. Similar analyses have been conducted in different settings, all on a country level (Heggeness & León, 2025).

These studies collectively point to a plausible mechanism: the burden of increased childcare responsibilities often fell disproportionately on mothers, leading to significant reductions in their labour force participation and hours worked. However, few studies focus on province-specific effects in British Columbia, where many analyses of country-wide phenomena may mask local effects. BC's partial childcare openness, with some centres remaining open, may have weakened disproportionate gender employment effects, as some parents may have been left untreated.

This paper addresses these province-specific gaps by analyzing Statistics Canada's Labour Force Survey microdata for British Columbia over 2019–2021, focusing on women's labour-market responses relative to men's during school and childcare closures. By employing a longer pre- and post-window and exploiting heterogeneity across child-age groups, this analysis estimates gendered impacts of closures. It complements national and international evidence while offering policy-relevant insights for BC's school and childcare systems.

3 Data & Limitations

This study utilizes public-use micro data from Statistics Canada's Labour Force Survey, a monthly cross-sectional household survey that collects detailed labour market information across Canada. The data are pooled from January 2019 to December 2021, providing sufficient pre- and post-COVID-19 periods to estimate changes in maternal employment following school and childcare disruptions.

The dataset has several limitations. Notably, the public-use file excludes identifiers sig-

nalling direct exposure to school or childcare closures[1], so treatment is inferred indirectly through proxies. The sample is restricted to parents whose youngest child is under age 17, as they were likely affected by these policies, while excluding parents with a youngest child aged 18 or older, who were presumably unaffected. This approach aligns with previous related studies (Qian & Fuller, 2020; Beauregard et al., 2022).

Additionally, the dataset lacks household identifiers, making it impossible to link individuals within households or account for regional linkages. This raises concerns about potential spillover effects, where one parent's employment might influence another's. However, controlling for economic family type (e.g., single earner family, dual earner family) partially mitigates this issue.

Before conducting the main analysis, the dataset was examined for missing values and other properties. Table 1 displays counts of missing observations and summary statistics for employment-related variables among parents. Summary statistics for categorical variables are omitted, as they do not provide meaningful interpretation; their proportions are instead reported in Table 2.

Table 1: Descriptive Statistics and Missing Values

Variable	Mean (SD)	Missing	25th Percentile	50th Percentile	75th Percentile
Employment	0.73 (0.44)	0	0.000	1.000	1.000
Female	0.53 (0.50)	0	0.000	1.000	1.000
Tenure with current employer (in months)	91.29 (75.47)	13564	26.000	69.000	145.000
Hours worked per week at main job	316.74 (173.45)	13564	220.000	360.000	400.000
5 year age group	–	0	–	–	–
Martial Status	–	0	–	–	–
Education	–	0	–	–	–
Class of Worker	–	9208	–	–	–
Industry	–	9208	–	–	–
Occupation	–	9208	–	–	–
Economic Family Type	–	0	–	–	–
Job permanency	–	26642	–	–	–
Full- or part-time status of last job	–	74886	–	–	–
Worked in the last year	–	65696	–	–	–
Immigration Status	–	0	–	–	–

Notes: Author's calculations using data taken from the LFS from 2019-2021 filtered to only include parents. Standard deviations are shown in parentheses. Hours worked may be scaled due to the dataset being public use. Summary statistics are not reported for categorical variables.

[1]The dataset does not include variables indicating direct exposure to childcare or school closures (e.g., daycare enrolment, homeschooling).

Table 2: Shares by Child Age Group and Gender

	Child <6		Child 6–12		Child 13–17	
	Male	Female	Male	Female	Male	Female
Labour Force Statistics						
Employed	84.38	52.60	83.95	71.04	80.08	72.91
(Standard Deviation)	(0.363)	(0.499)	(0.367)	(0.454)	(0.399)	(0.444)
Tenure with current employer in months	69.20	47.09	96.39	68.26	115.61	85.65
(Standard Deviation)	(64.308)	(56.763)	(80.375)	(74.272)	(92.848)	(85.529)
Hours worked per week at main job	339.10	156.21	336.69	223.37	323.13	237.69
(Standard Deviation)	(186.338)	(175.215)	(188.638)	(178.634)	(201.677)	(182.749)
Industry						
Management occupations	10.62	4.21	13.95	5.96	14.65	7.37
Business, finance and administration occupations, except management	7.40	17.54	6.67	20.63	6.75	19.50
Natural and applied sciences and related occupations, except management	11.51	3.95	11.83	2.95	8.86	2.30
Health occupations, except management	3.48	14.43	3.55	11.71	2.99	11.39
Occupations in education, law and social, community and government services, except management	7.89	14.52	8.30	17.84	8.76	17.23
Sales and service occupations, except management	11.27	16.06	11.52	18.38	11.25	19.84
Trades, transport and equipment operators and related occupations, except management	32.54	2.04	28.55	1.99	27.64	2.89
Natural resources, agriculture and related production occupations, except management	4.50	0.96	4.13	1.54	4.24	1.52
Occupations in manufacturing and utilities, except management	5.02	1.06	4.41	1.69	5.28	2.15
Other	3.21	22.59	4.79	14.41	8.04	13.57
Highest Education Level Attained						
Some high school	5.06	4.06	4.90	4.18	6.08	4.41
High school graduate	17.32	15.18	18.05	15.61	18.49	16.74
Some postsecondary	4.36	5.35	4.51	5.00	4.69	4.28
Postsecondary certificate or diploma	39.64	34.35	38.49	37.08	37.32	39.20
Bachelor's degree	22.39	27.81	20.84	25.17	20.25	22.65
Above bachelor's degree	10.60	12.66	12.41	12.33	11.91	11.71
Class of Worker						
Public sector employees	15.79	24.72	16.49	27.96	17.73	28.50
Private sector employees	62.00	42.28	55.70	43.05	49.29	43.94
Self-employed incorporated, with paid help	5.96	1.72	8.26	2.47	7.43	2.81
Self-employed incorporated, no paid help	4.52	1.70	5.72	2.27	6.65	2.34
Self-employed unincorporated, no paid help	7.00	6.46	7.19	8.80	9.05	7.64
Other	3.21	22.59	4.79	14.41	8.04	13.57
Economic Family Type						
Dual-earner couple, youngest child 0 to 17	64.86	59.44	68.81	59.96	67.47	56.84
Single-earner couple, man+ employed, youngest child 0 to 17	25.67	23.52	16.93	14.42	14.17	11.66
Single-earner couple, woman+ employed, youngest child 0 to 17	4.43	4.17	5.36	4.87	7.08	5.72
Non-earner couple, youngest child 0 to 17	2.96	2.72	2.37	2.04	3.48	2.89
Lone-parent family, parent employed, youngest child 0 to 17	1.66	6.33	5.46	14.63	6.04	18.28
Lone-parent family, parent not employed, youngest child 0 to 17	0.37	3.77	0.92	3.87	1.66	4.59
Immigrant Status						
Immigrant, landed 10 or less years earlier	12.80	15.47	7.00	7.59	4.86	5.25
Immigrant, landed more than 10 years earlier	14.96	12.55	21.52	21.10	24.86	26.86
Non-immigrant	72.24	71.98	71.49	71.31	70.28	67.89

Notes: Author's calculations using LFS data from 2019-2021. Percentages may not sum to 100 due to rounding or omission of variables that have few observations. Employment and categorical variables are multiplied by 100 to obtain percentages. Hours worked may be scaled due to the dataset being public use.

Table 1 shows that there are no missing values for Employment or Female. However, several employment-related variables, including class of worker, industry, occupation, tenure, and hours worked at the main job, do contain missing values. Dropping observations with missing values in these variables could create sample selection bias, since they are missing based on the dependent variable, and reduce the variation of interest, potentially leading to biased estimates. Variables capturing past employment status or permanent employment are excluded from the regression, as they are assumed to have no direct effect on current employment and are missing across all employment categories[2]. These missing values are assumed to be missing at random.

Thus, employment related variables, including class of worker, industry, occupation, tenure, and hours worked at main job are not dropped. The missing values for categorical variables (class of worker, industry, occupation) are given their own class, and tenure and hours worked at main job are coded as 0 for individuals who are not employed. They are also given an indicator that indicates if an individual was missing hours worked or tenure. This approach will likely increase variance, as there are large discrepancies for individuals across tenure and hours worked.

Table 2 reports descriptive statistics for key variables, including the the shares of observations for categorical variables, providing an overview of the sample. It shows that women have lower employment rates relative to men, with the gap narrowing for those with children aged 13–17 and widest for parents of children under the age of 6. Certain industries exhibit gender concentrations. For example, men are overrepresented in trades relative to women. Given industries' heterogeneous responses to school closures and COVID-19, this underscores the need for industry and occupation related controls.

In addition to a pooled model for all parents, analyses are conducted separately for parents whose youngest child is under 6, aged 6–12, or aged 13–17. Given the heterogeneous nature of school and childcare closures in British Columbia, the analysis uses separate subgroups. Specifically, daycares and preschools were not mandated to close, whereas K–12 institutions experienced province-wide shutdowns. Treatment intensity thus varies by youngest child's age, with age groups serving as exposure proxies, as parents bear less of a childcare burden with older children, and parents with a youngest child under the age of 6 may have been partially treated.

Table 3: Number of Parents by Children's Age and Gender

	Child < 6		Child 6–12		Child 13–17	
Year	Male	Female	Male	Female	Male	Female
2019	6220	6920	4776	5673	3102	3763
2020	4888	5389	4224	5053	2552	3139
2021	4415	4817	4130	4516	2641	3042
Total	15523	17126	13130	15242	8295	9944

Notes: Data taken from the Labour Force Survey from 2019–2021.

One concern with LFS data is measurement error; however, the main variables are un-

[2]Details on missing counts by employment are reported in Appendix A1.

likely to be misreported. Moreover, while weekly hours may be rounded and scaled, the large pooled sample size should prevent significant biases in aggregate total hours. Another potential issue is sample representativeness or gender imbalances. Table 3 reports gendered counts for each subgroup of interest. Across subgroups, women slightly outnumber men, but only by a minor margin. The sample for parents with teenagers is smaller than the others but remains substantial, with no major imbalances across gender. Thus, the final sample sizes are n=32,649 for parents with the youngest child under 6, n=28,372 for those aged 6–12, n=18,239 for those aged 13–17, and n=79,260 for parents overall. All analyses were carried out using R 4.4.1 (R Core Team, 2024).

4 Methodology

To estimate the gendered impacts of school closures, this study employs a difference-in-differences model. Although school closures represented a distinct policy shock, they coincided with the broader onset of the COVID-19 pandemic. Compared to other approaches, such as an OLS model, the DID model is appropriate because it isolates the effects of these closures on female employment status, separating them from other pandemic-related factors. In other words, the DID framework controls for time-varying confounders by comparing observed outcomes against a counterfactual represented by men. This choice is informed by existing literature, which consistently shows that women disproportionately bear childcare responsibilities. Moreover, Figure 1 reveals fairly similar pre-trends for women and men prior to school closures, with no evidence of divergence, confirming the appropriateness of the DID identification strategy.

School closures are defined as the months following April 2020, when British Columbia implemented full K–12 lockdowns. This timing reflects BC's declaration of closures on March 17, 2020; since LFS data is typically collected in the first week of each month, observations before April likely precede the closures. This approach aligns with prior COVID studies using LFS data (Singh et al., 2022). The model specification is as follows:

$$Y_{it} = \alpha + \gamma Female_i + \pi SchoolClosure_t + \delta(Female_i \times SchoolClosure_t) + \beta X_i + \lambda_t + \epsilon_{it} \quad (1)$$

Where Y_{it} represents employment status of individual i at time t (= 1 if employed; = 0 otherwise), $Female_i$ is a dummy variable and represents treatment (= 1 for female; = 0 otherwise), $SchoolClosure_t$ is a binary variable (= 0 for March 2020 and earlier; = 1 for April 2020 and later) capturing the full school closure period in British Columbia. X_i includes control variables such as 5 year age group (e.g., 15-19, 20-24), industry, occupation, marital status, economic family type (e.g., dual-earner couple, single-earner couple), class of worker (e.g., public sector, private sector), tenure with current employer, hours worked per week at main job, and immigration status. X_i also includes indicators for missing tenure with current employer and missing hours worked at main job since these values are missing for unemployed individuals. The use of these controls are justified from the share distribution in Table 2 which illustrate gendered differences in distribution. These controls are time invariant because the LFS is cross sectional, meaning each individual is not observed over

time. λ_t captures monthly fixed effects to account for seasonal patterns in employment, particularly relevant for industries such as agriculture and teaching. ϵ_{it} is the error term.

Robust standard errors address heteroskedasticity, and R automatically drops variables that are perfectly collinear. Although multicollinearity may increase the variance of the estimates, it does not bias coefficients and therefore does not threaten identification.

Equation 1 is estimated across four subgroups to account for heterogeneous treatment effects: all parents of children under 18; parents whose youngest child is under 6; parents whose youngest child is aged 6–12; and parents whose youngest child is aged 13–17. This reflects BC's lack of mandated daycare or preschool closures (potentially sparing some parents of young children from treatment) and the plausibly lower childcare responsibility for parents of teenagers relative to those with elementary school-aged children.

5 Results

Table 4: DID Estimates of School Closures on Employment

Dependent Variable:	Employment			
Model:	All	Child < 6	Child 6-12	Child 13-17
Variables				
Female	0.04384***	0.04168***	0.05475***	0.03979***
	(0.00255)	(0.00425)	(0.00406)	(0.00512)
SchoolClosure	0.00061	0.00045	0.00649*	-0.00395
	(0.00208)	(0.00332)	(0.00340)	(0.00439)
Female × SchoolClosure	-0.02217***	-0.02943***	-0.02841***	-0.00687
	(0.00300)	(0.00486)	(0.00478)	(0.00598)
Fixed-effects				
Month	Yes	Yes	Yes	Yes
Fit statistics				
Observations	79,260	32,649	28,372	18,239
R^2	0.77440	0.77981	0.77511	0.78335
Within R^2	0.77278	0.77925	0.77198	0.78059

Heteroskedasticity-robust standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Notes: Data taken from LFS from 2019-2021. Coefficients on the controls are not reported.

Table 4 presents the DID estimates of school closures on employment. The key coefficient of interest is the interaction term (Female × SchoolClosure), which captures the differential change in women's employment relative to men following the implementation of school closures. All regressions include month fixed effects and report heteroskedasticity-robust standard errors. Coefficients on control variables are not reported.

Column 1 ("All") reports results for all parents of children under 17. The coefficient on Female × SchoolClosure is -0.022, suggesting that, on average, mothers experienced a 2.2 percentage point decline in employment relative to fathers after school closures. This effect

is statistically significant at the 1% level, suggesting that the probability of obtaining a result as extreme or more than the one observed is less than 1%.

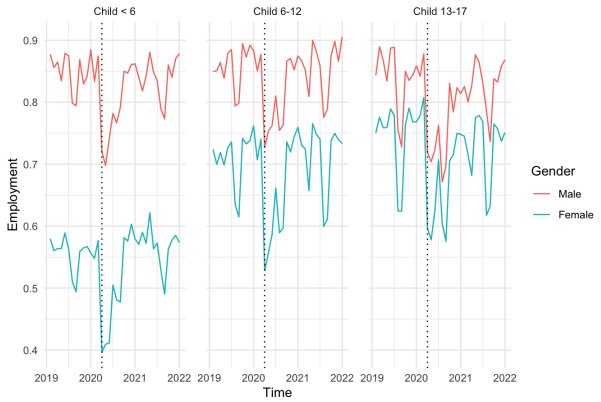
To examine the differential impacts this had on different parents, the same model is estimated for three subgroups defined by the age of the youngest child. Column 2 ("Child $<$ 6") includes parents whose youngest child is under 6. This coefficient is the largest in magnitude and suggests that, mothers on average experienced a 2.9 percentage point decline in employment relative to men, significant at the 1% level. Column 3 ("Child 6–12") corresponds to parents of children in elementary school. The estimated effect is -0.028, also significant at the 1% level, suggesting that mothers with school-aged children, on average, faced a 2.8 percentage point decline in employment relative to men. Column 4 ("Child 13–17") includes parents of teenagers. The interaction coefficient is -0.007 and statistically insignificant, suggesting minimal change in employment for mothers, relative to men, of older children.

These findings align with theoretical expectations that mothers of young children would bear the largest childcare burden. Consequently, mothers of teenagers experienced the smallest change in employment, as their children are likely independent enough to require little constant supervision.

One key assumption underlying the validity of the difference-in-differences approach is the parallel trends assumption, which is subseqeuntly verified by regressing the outcome variable, employment, on the list of control variables. Although the main analysis employs heteroskedasticity-robust standard errors, occupation-clustered standard errors are used as a robustness check to account for potential correlations in employment outcomes within industries. Additionally, a secondary analysis examines whether fully online learning led to the largest declines in maternal employment, given the varying intensities of school closures; these estimates will be compared to the main findings below.

5.1 Parallel Trends

Figure 1: Parallel Trends



Notes: Data taken from LFS from 2019–2021. Dotted line represents April 2020.

Figure 1 provides a visual check of the parallel trends assumption. It plots the residuals from regressing employment on the control variables, with a dotted line marking the school closure date in April 2020. The red line represents fathers and the blue line represents mothers. As shown, pre-closure trends for women and men are similar across all subgroups, exhibiting no evident divergence. This supports the parallel trends assumption, validating the use of men as a counterfactual for women and the difference-in-differences approach.

5.2 Clustered Standard Errors

Equation 1 is re-estimated using occupation-clustered standard errors to account for potential correlation in employment outcomes within industries. The results are presented in Table 5.

Table 5: Occupation Clustered DID Estimates of School Closures on Employment

Dependent Variable:	Employment			
Model:	All	Child < 6	Child 6-12	Child 13-17
Variables				
Female	0.03268***	0.02547**	0.04938***	0.04022***
	(0.00791)	(0.01148)	(0.00957)	(0.01075)
SchoolClosure	-0.00048	-0.00167	0.00618	-0.00274
	(0.00348)	(0.00566)	(0.00606)	(0.00840)
Female × SchoolClosure	-0.02218***	-0.02747**	-0.03168***	-0.00907
	(0.00636)	(0.01049)	(0.00779)	(0.00927)
Fixed-effects				
Month	Yes	Yes	Yes	Yes
Fit statistics				
Observations	79,260	32,649	28,372	18,239
R^2	0.74294	0.75759	0.73553	0.74366
Within R^2	0.74110	0.75697	0.73185	0.74039

Clustered (occupation) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Notes: Data taken from LFS from 2019-2021. Coefficients on the controls are not reported. Standard errors are obtained from 44 occupation clusters.

The key interaction term (Female × SchoolClosure) remains negative across all subgroups, indicating that school closures were associated with declines in mothers' employment relative to fathers. For all parents, the coefficient is -0.022 ($p < 0.01$), nearly identical to the heteroskedasticity-robust estimate from Table 4, implying a 2.2 percentage point decline in employment for mothers relative to men.

Subgroup estimates also remain consistent with the main results. Mothers with a youngest child under 6 show a -0.027 decline (compared to a -0.029 previously), those with children aged 6–12 show a -0.032 decline (vs. -0.028), and mothers of teenagers show a small, statistically insignificant decline of -0.009 (vs. -0.007). As before, the largest employment losses occur for mothers of elementary school-aged children, consistent with higher childcare demands, while those with teenage children remain statistically insignificant.

Although the significance levels differ slightly between the heteroskedasticity-robust and clustered specifications, the point estimates for children under 6 and ages 6–12 remain similar across models. Given their close magnitudes, these estimates indicate that mothers with younger and school-aged children experienced the largest employment declines relative to fathers. The apparent change in which subgroup has the largest effect arises from differences in standard-error precision rather than meaningful differences in underlying effect sizes. Overall, the clustered estimates closely mirror the main findings, suggesting that the results are robust to the choice of variance estimator.

5.3 Extension: Heterogeneity by Treatment Intensity

To investigate when employment changes were most substantial, Equation 1 is modified so that $SchoolClosure_t$ takes on three values (= 0 for March 2020 and earlier; = 1 for April–May 2020; = 2 for June 2020 and later). This coding reflects the phased reopening of schools and hybrid learning after June 1, 2020, which may have reduced the childcare burden in later periods.

Table 6: DID Estimates of Heterogeneous School Closures on Employment

Dependent Variable:	Employment			
Model:	All	Child < 6	Child 6-12	Child 13-17
Variables				
Female	0.04346***	0.04125***	0.05442***	0.03952***
	(0.00255)	(0.00425)	(0.00405)	(0.00511)
SchoolClosure1	-0.01978***	-0.01422**	-0.01861**	-0.03015***
	(0.00451)	(0.00692)	(0.00748)	(0.01004)
SchoolClosure2	0.00349	0.00274	0.00972***	-0.00030
	(0.00213)	(0.00344)	(0.00348)	(0.00446)
Female × SchoolClosure1	-0.05000***	-0.05539***	-0.06032***	-0.02721**
	(0.00624)	(0.00941)	(0.01059)	(0.01319)
Female × SchoolClosure2	-0.01691***	-0.02409***	-0.02277***	-0.00308
	(0.00308)	(0.00506)	(0.00487)	(0.00609)
Fixed-effects				
Month	Yes	Yes	Yes	Yes
Fit statistics				
Observations	79,260	32,649	28,372	18,239
R^2	0.77493	0.78016	0.77590	0.78393
Within R^2	0.77332	0.77960	0.77279	0.78118

Heteroskedasticity-robust standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Notes: Data taken from LFS from 2019-2021. Coefficients on the controls are not reported. Occupation-clustered standard errors are reported in Appendix A2.

Table 6 reports the results using heteroskedasticity-robust standard errors. Results using occupation clustered standard errors are reported in Appendix A2. The coefficients of

interest are the interactions between Female and the school closure indicators, Female × SchoolClosure1 (fully remote learning) and Female × SchoolClosure2 (hybrid reopening).

Across all parents, Female × SchoolClosure1 is -0.050, suggesting that mothers experienced a 5 percentage point decline in employment relative to fathers during the fully remote learning period, significant at the 1% level. Similar to Table 4, subgroup analyses reveal heterogeneous effects: mothers with children under 6 experienced a 5.5 percentage point decline, mothers of children aged 6–12 experienced the largest decline at 6.0 percentage points, and mothers of teenagers experienced a smaller but still negative decline of 2.7 percentage points. All coefficients are statistically significant ($p < 0.05$).

These results suggest that employment effects were most pronounced during the period of fully remote learning and were largest for mothers of elementary school-aged children, consistent with the higher childcare demands associated with this group. The smaller effects observed during the partial reopening period indicate that hybrid or part-time schooling may have partially alleviated the childcare burden.

Overall, aligning with previous studies (Qian & Fuller, 2020; Singh et al., 2022), the results suggest that school closures disproportionately reduced mothers' employment, on average, relative to fathers, with the largest effects observed for mothers of young children. The main DID estimates show a 2–3 percentage point decline across subgroups, and these findings remain robust when standard errors are clustered by occupation, confirming that the results are not sensitive to assumptions imposed on the variance estimator.

The extension exploiting variation in treatment intensity reveals that employment declines were most pronounced during periods of fully remote learning, with hybrid or part-time reopenings partially offset these effects. Across all specifications, mothers of teenagers experienced minimal employment losses.

6 Conclusion

This paper employs a difference-in-differences approach with 2019–2021 Canadian Labour Force Survey data to assess how COVID-19 school closures affected mothers' employment. Across specifications, school closures consistently reduced mothers' employment more than fathers', with the largest declines among mothers of elementary school-aged children, followed by those with children under six, and minimal effects for mothers of teenagers. These patterns are robust to heteroskedasticity-robust and occupation-clustered standard errors and are strongest during full school closures, underscoring the role of increased childcare demands. While data constraints limit the analysis, they are unlikely to overturn the central findings.

The findings have clear real-world implications. They suggest that pandemic-related school closures amplified existing gender disparities in the labour market and highlight the importance of childcare accessibility in supporting maternal employment. Policies aimed at mitigating the employment costs of school closures—such as flexible work arrangements, subsidized childcare, or targeted support for parents of young and school-aged children—could help reduce these disparities in future crises.

Future research could explore longer-term labour market consequences for mothers, including potential impacts on career progression, earnings, and part-time versus full-time employment. Additionally, methodological improvements, such as household and regional

indicators, would greatly strengthen the results obtained in this paper.

7 Appendix

Table A1: Missing Values for Each Group

Variable	Employed	Unemployed
Class of Worker	0	9208
Occupation	0	9208
Industry	0	9208
Permanent or Temporary Status	11621	15021
Full- or part-time status of last job	57850	17036
Worked in the last year	57850	7846
Tenure with current employer (in months)	0	13564
Hours worked per week at main job	0	13564

Notes: Data taken from LFS from 2019–2021.

Table A2: DID Estimates of Heterogeneous School Closures on Employment (Occupation clustered SE)

Dependent Variable:	Employment			
Model:	All	Child < 6	Child 6-12	Child 13-17
Variables				
Female	0.04346***	0.04125***	0.05442***	0.03952***
	(0.00671)	(0.01016)	(0.00784)	(0.01018)
SchoolClosure1	-0.01978***	-0.01422	-0.01861*	-0.03015**
	(0.00652)	(0.01115)	(0.00966)	(0.01262)
SchoolClosure2	0.00349	0.00274	0.00972*	-0.00030
	(0.00298)	(0.00459)	(0.00494)	(0.00797)
Female × SchoolClosure1	-0.05000***	-0.05539***	-0.06032***	-0.02721
	(0.01373)	(0.01883)	(0.01629)	(0.01630)
Female × SchoolClosure2	-0.01691***	-0.02409***	-0.02277***	-0.00308
	(0.00516)	(0.00820)	(0.00628)	(0.00879)
Fixed-effects				
Month	Yes	Yes	Yes	Yes
Fit statistics				
Observations	79,260	32,649	28,372	18,239
R^2	0.77493	0.78016	0.77590	0.78393
Within R^2	0.77332	0.77960	0.77279	0.78118

Clustered (occupation) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Notes: Data taken from LFS from 2019-2021. Coefficients on the controls are not reported. Standard errors are obtained from 44 occupation clusters.

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